

DSM Mini Project

Topic:- Dynamic modelling of a Ball Balancing Stewart platform.

Forces and Energy equations/assumptions:-

1) Forces acting on the ball:-

Gravitational force: mg

Normal reactions from platform

Frictional force (rolling/sliding resistance)

2) Equation of motion - (x axis)

when angle is tilted by angle α

$$m \frac{d^2 x}{dt^2} = mg \sin(\alpha)$$

Using small-angle approximation ($\sin \alpha \approx \alpha$)

$$\frac{d^2 x}{dt^2} = g \alpha$$

3) Equation of motion - (y axis)

similarly for tilt angle β

$$\frac{d^2 y}{dt^2} = g \beta$$

4) Control input (PID controller):-

$$\alpha = K_p e_x + K_i \int e_x dt + K_d \frac{de_x}{dt}$$

$$\beta = -(K_p e_y + K_i \int e_y dt + K_d \frac{de_y}{dt})$$

5) Final Governing equations:-

$$\frac{d^2 x}{dt^2} = g(K_1 e_y + K_2 \int e_y dt + K_3 \frac{de_y}{dt})$$

$$\frac{d^2 y}{dt^2} = -g(K_1 e_x + K_2 \int e_x dt + K_3 \frac{de_x}{dt})$$

6) Kinetic Energy:-

$$K.E = \frac{1}{2} m (v_x^2 + v_y^2 + v_z^2)$$

7) Potential Energy:-

$$P.E = mgh$$

8) Energy loss (Friction):-

$$E_{\text{loss}} = \int F_{\text{friction}} \cdot v dt$$

9) Steady state Condition:-

At equilibrium:

- Ball velocity = 0
- Ball position = centre
- Control error $\rightarrow 0$

10) State Space Representation

let

$$x_1 = x, \quad x_2 = \frac{dx}{dt}$$

$$x_1' = x_2$$

$$x_2' = g \cos \alpha$$

This can be written as:

$$\dot{X} = AX + BU$$

11) Units Check:-

Position $\rightarrow$ meters (m)

Velocity $\rightarrow$ m/s

Acceleration $\rightarrow$ m/s²

Force $\rightarrow$ Newton (N)

Energy $\rightarrow$ Joules (J)

All equations are dynamically consistent.

12) Physical Interpretation:-

Ball accelerates in direction of platform tilt

System is second-order and unstable without control.

PID controls stabilize the system at equilibrium.

Energy decreases due to friction $\rightarrow$ realistic behaviour

13) Key assumptions:-

Platform is a rigid body

Small angle approximation is valid.

Falling with limited slip.

Ideal actuators (no delay)

Uniform gravitational field.

No air resistance